

MATH 118 Homework Assignment 4

Q1) $N = 5000 \text{ psi}$ $\sigma = 400 \text{ psi}$ $n = 36$

$$P(4800 < \bar{X} < 5200) = ?$$

① Since the sample size ($n=36$) is greater than 30, the CLT tells us that the distribution of the sample mean ($\bar{X}$) is approximately Normal.

$$\mu_{\bar{X}} = N = 5000$$

$\sigma_{\bar{X}}$ standard error calculates by $\sigma_{\bar{X}} = \frac{\sigma}{\sqrt{n}}$

$$\sigma_{\bar{X}} = \frac{400}{\sqrt{36}} = \frac{400}{6} \approx 66.67$$

$$Z = \frac{\bar{X} - \mu_{\bar{X}}}{\sigma_{\bar{X}}}$$

$$Z_{\text{lower}} = \frac{4800 - 5000}{66.67} = \frac{-200}{66.67} \approx -3.00$$

$$Z_{\text{upper}} = \frac{5200 - 5000}{66.67} = \frac{200}{66.67} \approx 3.00$$

$$Z_{\text{score}} = Z_{\text{upper}} - Z_{\text{lower}} = P(-3.00 < Z < 3.00) = 0.9987 - 0.0013$$

$$Z_{\text{score}} = 0.9974 = P(-3.00 < Z < 3.00)$$

$$\textcircled{2} P(4900 < \bar{X} < 5100) = 0.99 ?$$

The remainin probability is $1 - 0.99 = 0.01$

This split equally into two tails $= 0.01/2 = 0.005$

$$Z_{\text{upper limit}} \text{ is } = 0.99 + 0.005 = 0.995 \Rightarrow Z = 2.575$$

$$Z = 2.575$$

$$Z = \frac{\bar{X} - \mu}{\sigma_{\bar{X}}} \Rightarrow 2.575 = \frac{5100 - 5000}{\sigma_{\bar{X}}}$$

$$\sigma_{\bar{X}} \approx 38.82$$

$$\sigma_{\bar{X}} = \frac{\sigma}{\sqrt{n}} = \frac{400}{\sqrt{n}} = 38.82$$

$$\Rightarrow \sqrt{n} \approx 10.304 \Rightarrow n \approx 106.17$$

$$\Rightarrow n = 107$$

Q2) $n=16$ σ is unknown $\bar{X}=50.3 \text{ mm}$ $S=0.8 \text{ mm}$

Q.1)
CL = 90.95

Since σ is unknown and $n < 30$, we use t -distribution

$$df = n - 1 = 16 - 1 = 15 = df$$

critical t -value ($\alpha/2$)

for a 90.95 confidence level the significance level is $\alpha = 1 - 0.95 = 0.05$

We need to find $t_{0.05/2} = t_{0.025}$ with $df=15$

from t -table with $df=15$;

$$t_{0.025, 15} = 2.131$$

SE (standard error)

$$SE = \frac{S}{\sqrt{n}}$$

$$SE = \frac{0.8}{\sqrt{16}} = 0.2 \text{ mm}$$

ME (Margin of Error)

$$ME = t_{\alpha/2} \cdot SE$$

$$ME = 2.131 \times 0.2 = 0.4262$$

CI (Confidence Interval)

$$CI = \bar{X} \pm ME$$

$$CI = 50.3 \pm 0.4262$$

$$\text{Lower bound} = 50.3 - 0.4262 = 49.8738 \text{ mm}$$

$$\text{upper bound} = 50.3 + 0.4262 = 50.7262 \text{ mm}$$

The 90.95 confidence interval for the population mean length (μ) of the metal rods is:

$$49.8738 \text{ mm} < \mu < 50.7262 \text{ mm}$$

2.2 For the t -confidence Interval constructed for μ to be valid and reliable, the following two primary assumptions must be met:

1) Simple Random Sample (Independence)

The sample of $n=16$ rods must be a simple random sample (SRS) drawn from the population, and the measurements must be independent of each other.

• Rationale: This ensures that the sample is representative of the entire population of rods and prevents bias in the estimate of μ .

2) Normality of The Population

The population distribution of the rod lengths must be approximately Normal.

• Rationale: Since the sample size ($n=16$) is small ($n < 30$) and the population standard deviation (σ) is unknown, the use of the t -distribution is necessary. The t -distribution requires the underlying population to be normal for the confidence interval calculations to be accurate.

2.3 Critical-Z-value

$$ME = 0.2$$

$$CL = 0.95, \alpha = 0.05$$

$$z_{\alpha/2} = z_{0.025}$$

$$z_{0.025} = 1.96$$

$$ME = z_{\alpha/2} \cdot \frac{s}{\sqrt{n}}$$

$$0.2 = 1.96 \cdot \frac{0.8}{\sqrt{n}} \Rightarrow \sqrt{n} = \frac{(1.96)(0.8)}{0.2}$$

$$\sqrt{n} = 7.84$$

$$n = (7.84)^2 \approx 61.4656 \Rightarrow n = 62$$

$$Q3) \hat{\mu}_1 = \bar{X} = \frac{1}{n} \sum_{i=1}^n x_i$$

$$(31) E[\hat{\mu}_1]$$

$$E[\hat{\mu}_1] = E\left[\frac{1}{n} \sum_{i=1}^n x_i\right]$$

Since $E[x_i] = \mu$ for all i ;

$$E[\hat{\mu}_1] = \frac{1}{n} \sum_{i=1}^n \mu$$

$$E[\hat{\mu}_1] = \frac{1}{n} (n\mu) = \mu$$

$$\text{Bias}(\hat{\mu}_1) = E[\hat{\mu}_1] - \mu = \mu - \mu = 0$$

Since $\text{Bias}(\hat{\mu}_1) = 0$, the sample mean $\bar{X}$ is an unbiased estimator of μ .

$$\hat{\mu}_2 = \frac{x_1 + 2x_2}{3}$$

$$E[\hat{\mu}_2] = E\left[\frac{1}{3}x_1 + \frac{2}{3}x_2\right], E[x_1] = \mu \text{ and } E[x_2] = \mu$$

$$E[\hat{\mu}_2] = \frac{1}{3}\mu + \frac{2}{3}\mu = \mu$$

$$\text{Bias}(\hat{\mu}_2) = E[\hat{\mu}_2] - \mu = \mu - \mu = 0$$

Since $\text{Bias}(\hat{\mu}_2) = 0$ the estimator $\hat{\mu}_2$ is also an unbiased estimator of μ .

$$\mu_1 = \bar{X} \quad E[\hat{\mu}_1] = \mu \quad \text{Bias}(\mu) = 0 \quad \text{unbiased}$$

$$\mu_2 = \frac{x_1 + 2x_2}{3} \quad E[\hat{\mu}_2] = \mu \quad \text{Bias}(\mu) = 0 \quad \text{unbiased}$$

$$(32) \text{Var}(\hat{\mu}_1) = \text{Var}\left(\frac{1}{n} \sum_{i=1}^n x_i\right)$$

$$\text{Var}(\hat{\mu}_1) = \left(\frac{1}{n}\right)^2 \text{Var}\left(\sum_{i=1}^n x_i\right) \text{ since } \text{Var}(x_i) = \sigma^2 \text{ for all } i$$

$$\boxed{\text{Var}(\hat{\mu}_1) = \frac{1}{n^2} n\sigma^2 = \frac{\sigma^2}{n}}$$

$$\text{Var}(\hat{\mu}_2) = \text{Var}\left(\frac{1}{3}x_1 + \frac{2}{3}x_2\right) \Rightarrow \text{Since } \text{Var}(x_1) \text{ and } \text{Var}(x_2) = \sigma^2$$

$$\text{Var}(\hat{\mu}_2) = \left(\frac{1}{3}\right)^2 \sigma^2 + \left(\frac{2}{3}\right)^2 \sigma^2 = \frac{5}{9} \sigma^2$$

Estimator	Var
$\hat{\mu}_1 = \bar{X}$	$\frac{\sigma^2}{n}$
$\hat{\mu}_2 = \frac{x_1 + 2x_2}{3}$	$\frac{5\sigma^2}{9}$

3.3 $MSE(\hat{\theta}) = \text{Var}(\hat{\theta}) + (\text{Bias}(\hat{\theta}))^2$

$\hat{\mu}_1 = \bar{X}$

$$MSE(\hat{\mu}_1) = \text{Var}(\hat{\mu}_1) + (\text{Bias}(\hat{\mu}_1))^2$$

$$MSE(\hat{\mu}_1) = \left(\frac{\sigma^2}{n}\right) + 0$$

$$MSE(\hat{\mu}_1) = \frac{\sigma^2}{n}$$

$\hat{\mu}_2 = \frac{x_1 + 2x_2}{3}$

$$MSE(\hat{\mu}_2) = \text{Var}(\hat{\mu}_2) + (\text{Bias}(\hat{\mu}_2))^2$$

$$MSE(\hat{\mu}_2) = \frac{5\sigma^2}{9} + 0$$

$$MSE(\hat{\mu}_2) = \frac{5\sigma^2}{9}$$

3.4

prefer $\hat{\mu}_1 = \bar{X}$ (the sample mean)

The choice is based on the Mean Squared Error (MSE) where the estimator with the smallest MSE is preferred.

$$MSE(\hat{\mu}_1) = \frac{\sigma^2}{n}$$

$$MSE(\hat{\mu}_2) = \frac{5\sigma^2}{9} \approx 0.556\sigma^2$$

① Lower MSE: for any practical sample size $n \geq 2$, $MSE(\hat{\mu}_1)$ is smaller than $MSE(\hat{\mu}_2)$

② $\hat{\mu}_1$ is a consistent estimator. As the sample size n increases by collecting more data, $MSE(\hat{\mu}_1)$ approaches 0. $\hat{\mu}_2$, however, ignores most of the data and its MSE remains fixed at $0.556\sigma^2$, regardless of n .

Therefore, $\hat{\mu}_1$ is a more efficient and reliable estimator.